Every trial is an exercise in reasoning under uncertainty. A jury weighs a smear of DNA against an alibi; a commercial judge asks what an insurer would have done had it been told the truth; an overloaded court must decide which of millions of cases to hear first. Probability is the language of the first problem and expected utility the language of the second. Bayes' rule joins them.

Bayes and Law is a casebook for lawyers, statisticians and policy makers. It works through the classic problems, from the island problem and the O. J. Simpson battering argument to *People v Collins*, *R v Adams* and Sally Clark; reads English tort, misrepresentation and contract law as Bayesian constraints, with the collapse of Greensill Capital as a case study; turns stories and pleadings into generative models; and draws on the work of Lindley, Fienberg, Kadane, Dawid, Mortera, Smith, Balding and Donnelly. It compares English and French law, tests the legal origins thesis in Africa, models India's court backlog as a queue, follows Gould into Corsica and Gambetta into Sicily, and applies Becker and Murphy to costs and deterrence.

The final part asks Lindley's question: should machines decide? It explains how Lean style verification, explainable AI and Bayesian standards of calibration can make legal decisions checkable, and why a court that weighs odds rather than opinions serves justice better.

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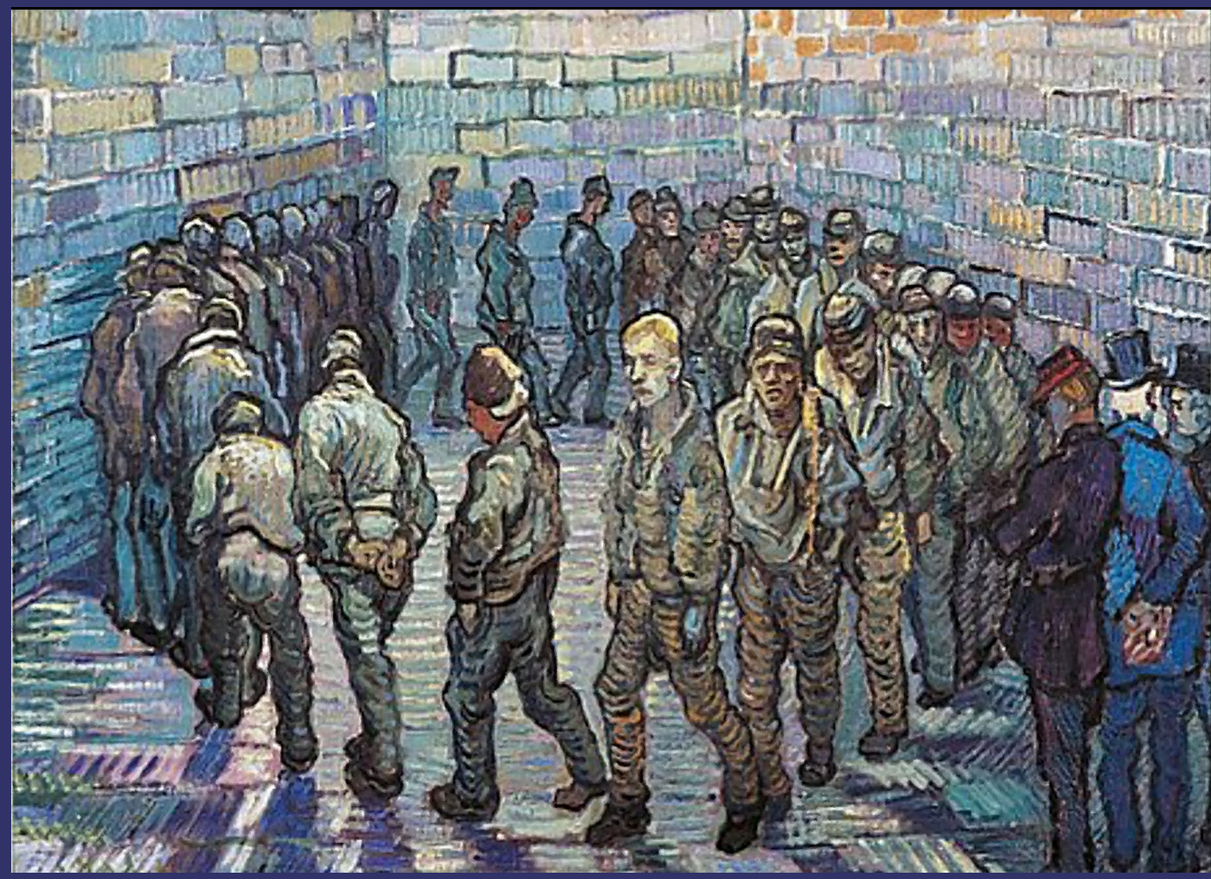
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Bayes and Law

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Evidence, Expected Utility and Machines in the Courtroom



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